Project Proposal

Executive Summary

A new multilingual voice banking assistant designed to help users access their banking services by seamlessly interacting with the assistant in a natural, spoken manner. The Banking Assistant will provide users with an opportunity to obtain accurate, spoken banking responses through free conversation in their preferred language, as opposed to requiring them to navigate through menus found within many traditional IVR systems.

Utilizing the latest advancements in technology, the Banking Assistant will combine modern-day speech processing and LLM (Large Language Models) technologies together to create an intelligent, easy-to-use and scalable voice interface for banking.

The Banking Assistant will consist of three primary components: the Speech Processing Layer, the Language Intelligence Layer and the Response Generation Layer. The assistants' Speech to Text and Text to Speech models (e.g., Sarvam) will provide support for Indian voice input and multilingual output. The Language Intelligence Layer will utilize LLM to process the user's question, provide context for the intent and understand the question being asked. The Response Generation Layer will also use LLM to create a simple, clear, relevant reply based on the user's identified intent and service request.

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The voice banking assistant will support five major features: onboarding users with voice-based registration and login; assisting users with querying general banking inquiries; providing users the ability to access their bank account balance; allowing users to view their recent bank transaction history; and providing users with information about various schemes and privileges offered by the bank. Each of these features has been designed to help both the information gathering and account management functions of the banking experience and illustrate how voice AI can help to improve daily banking interactions.

The project's goal is to build a voice AI prototype that improves the experience of non-English speakers, seniors, and people in technologically isolated or poorly served communities by enabling multi-lingual, real-time banking. The result of this work is a working prototype that demonstrates how voice AI can enhance the customer experience, decrease dependence on traditional IVR systems, and be the basis for the creation of the next generation of AI-powered banking platforms.

1 Background

The way in which people bank has changed dramatically due to the speed at which banks have been digitising their banking products. Smartphones, the internet and the use of automated customer service technology means that many of the transactions people perform on a daily basis now take place through a digital channel. There is still a considerable gap in terms of accessibility to these products for a large segment of the population. In particular, users in some countries (I.e., India) that have multiple languages other than English, the vast majority of digital banking applications and online

banking interfaces are predominantly written in English and present challenges to most users. This gap is even bigger among certain sub-populations - including those who are older, those who live in rural areas and those who are new to digital banking.

Typically, banks rely on the IVR (Interactive Voice Response) system for phone-based support. While IVR's assist in managing a bank's call centre by providing phone-based support to customers, IVR's are also inflexible and typically provide only a menu of language(s) and options to customers. Customers are forced to navigate through a series of predefined options which can create frustration, result in longer phone calls and produce customers with unsatisfied issues. Lack of flexibility results in IVR systems lacking the ability to "understand" and use natural language, answer complex questions and adapt dynamically to customer needs.

With advancements in speech recognition, the development of text-to-speech engines and the development of large-scale conversational duck models, conversational voice systems offer enormous opportunities to create newer models that improve user experiences with voice systems. Today's speech recognition models offer the ability to accurately transcribe users' speech in regional accents and dialects and large-scale conversational duck models have the ability to interpret and infer customers' intent(s) and context through the language of the conversation.

In conjunction with creating a more human-centered alternative to traditional IVRs, the technological improvements will support a transformative model of Voice AI for multilingual banking by demonstrating that conversational voice interfaces can provide a seamless experience for users. When used in the

context of a multilingual banking environment, such as this project, Voice AI encompasses many different technologies, including a conversational platform that enables customers to communicate via voice to receive answers, assistance, and guidance regarding their banking services. This enables users to communicate in their native languages; as a result, there is less of a learning curve when using digital banking services. Therefore, access to financial information/services will be greatly enhanced through this model. The project will explore how the integration of Speech Models (e.g., Sarvam) with Large Language Model (LLM)-based understanding and response generation can support creating a better model for Voice AI in the banking industry. This is expected to further demonstrate how this combination of technologies can create a transformational model for how banks and other financial institutions support their customers in a more human-centered way.

2 Statement of the problem

Voice-based banking systems currently exist in the form of IVR (Interactive Voice Response) systems which provide limited customer service due to their rigid and menu-based nature. They do not provide a means for customers to hold natural conversations with the banking institution nor do they work well with local dialects, regional languages or varying accents. They are also ineffective for very complicated or open-ended inquiries.

Customers often experience high levels of frustration and longer wait times due to their need to navigate through multiple menu options just to connect to a person. Consequently, there is an urgent need for a solution that uses voice-enabled banking with the capability to understand natural language, support

multiple languages, and produce intelligent, context-aware responses based on the customer's request.

The lack of these capabilities prevents accessibility of banking services to non-English speaking individuals, older users who may not be comfortable with technology, and those citizens who have not been afforded opportunities to learn digital technology. This initiative aims to resolve this issue by developing an LLM-powered, multilingual voice assistant capable of carrying out authentic banking transactions via natural speech.

3 Objectives

Develop and create a Multilingual Voice Banking Assistant.

Integrate Speech-to-Text and Text-to-Speech technologies for Natural Voice Interaction.

Implement LLM for Intent Detection and Context Recognition to understand user LLM questions.

Implement LLM to Generate Accurate and Conversational Responses to Banking Services.

Provide through the voice banking assistant the following core features:

Voice-enabled creation of accounts (Log in and Register).

General Assistance with Banking.

Query Your Bank Account Balances.

Reviewing Bank Account Transaction History.

Reviewing Bank Benefit Programs and Offers.

Build a Modular and Scalable Architecture for Production-Ready Banking Applications.

4 Deliverables

Completely Integrated Multilingual Voice AI Prototype

Speech-to-Text and Text-to-Speech Modules

LLM Component for Detecting Intent and Context

LLM Module for Producing an Accurate and Conversational Response to Banking Service Requests.

Onboarding, General Banking Request Handling, and All Other Account Functionality Modules

Documentation that describes the Architecture of the Entire System

A Report That Evaluates and Tests System Performance

Final Demonstration of the Voice Banking Agent.

5 Project Plan with Major Milestones

- All projects must be delivered within a specific timeframe as specified in the call.
- The major milestones to be achieved should be specified clearly in the project plan.
- Monthly milestones can be specified in the project plan.

Sr. No.	Milestone	Target Date	Remarks
1	**Data Annotation & Enhancement** – Expert validation, apply augmentation & labeling	Month 2, day 25	Complete dataset (~5,000+) ready, balanced across diseases and healthy classes
2	**Baseline Model Training & Model Optimization**	Month 6, day 30	First working ML model established with baseline performance metrics
3	**System Testing & Validation** – Controlled environment evaluation; refine AI/IoT integration	Month 7, day 20	
4	**Field Trials & Farmer Feedback** – Pilot testing on	Month 8, day 30	

	farms, farmer interaction, usability study		

6 Resources and budget

Resources Required for Calculating and Building Cloud Infrastructure

Utilization of GPU and/or Cloud Hosting Resources for running speech engines, Large Language Model APIs and Backend Services as well as for testing all elements involved in a voice pipeline (i.e., Google CoLab and/or Kaggle, or by using a low-cost VM Cloud).

Approximate Cost: ₹1500

APIs for speech processing and LLM's Language

Limited use of Sarvam (or an equivalent) for the purposes of Speech Recognition and Text to Speech, as well as other Hosted LLM solutions, including Gemini and/or Open-source LLMs, during the Development and Testing of these models in relation to accurately interpreting and responding to customer queries.

Approximate Cost: ₹3000

Stack of Software Tools for the Development of these models

Python, FastAPI/Flask, PyTorch/TensorFlow, Library of Speech Models, Front-end Frameworks. All of which are predominantly Open-Source and utilized to build the AI Pipeline, Backend Interface, and Integration Capability of the Models.

Approximate Cost: ₹0/- (Open-source)

Developing a Working Sample/Prototype (Web/Mobile Voice Application)

Creation of a functional prototype/sample application (Web Application / Mobile Application) that has the capabilities of processing voice input/output, providing an Authentication Flow and providing a demonstration of the Banking feature. This will include hosting of the application and the conducting of user testing and the actual deployment of the application.

Approximate Cost: ₹4000

Testing, Tuning and Documenting

Conducting tests across Multiple Languages, optimizing the models for performance, evaluating the models and preparing Reports and Conducting Final Demonstrations regarding the Project.

Approximate Cost: ₹1500

7 Appendix

Overview of System Architecture:

This system is composed of three layers in a modular design.

First layer - Speech Processing Layer (i.e. Speech-to-Text (STT) and Text-to-Speech (TTS) conversion). This will enable customers to communicate with the banking virtual assistant (VA) in their native language and dialect, thus enabling a high quality experience.

Second layer - Language Intelligence Layer. This will use LLMS to provide context about user intent based on a query regarding the services provided by the bank.

Third layer - Response Generation Layer. This will generate a response based on the intent of the user service request, with the response being clear, concise, and conversational.

Supported Banking Use Cases:

This prototype supports several Banking Use Cases, including:

1. Voice Registration/Login
2. General Queries about Banking
3. Queries about Bank Balance
4. Review of recent transaction history

Tools & Technologies Used:

1. Programming language = Python
2. Backend Framework = FastAPI/Flask
3. AI/ML Framework = PyTorch/TensorFlow
4. Speech Models = Sarvam (or comparable model)
5. LLMs = Gemini or other open-source LLM provider
6. Development Platforms = Google Colab /Kaggle
7. Deployment = Low cost Cloud VM
8. Banks Scheme, offer and benefit Information.

Team Members

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